

# DEVONTAE C. BAXTER

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## RESEARCH IDENTITY

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### Galaxy Evolution and the Cosmic Web:

- Performing spectroscopic surveys to connect galaxies with their local and large-scale cosmic web environments;
- Constraining the drivers of differential galaxy evolution using multi-wavelength imaging and spectroscopy.

### Computational Astrophysics and Astronomical Data Science:

- Modeling the physics driving environmental quenching across cosmic time.
- Applying machine learning methods to establish predictions for next-generation extragalactic surveys.
- Probing the formation and evolution of rare extragalactic objects (e.g., galaxy protoclusters, & isolated quenched dwarfs).

### Active Collaborations:

- Charting Cluster Construction with VUDS and ORELSE (C3VO);
- Gemini Observations of Galaxies in Rich Early ENvironments (GOGREEN);
- The Grism Reionization And Cosmic Evolution Survey (GRACE);

**Unified Astronomy Thesaurus Keywords:** Galaxy evolution, Galaxy quenching, High-redshift galaxy clusters.

## EMPLOYMENT

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**Future Faculty in the Physical Sciences (FFPS) Fellow at Princeton University** **Aug 2026 - present**

*Mentor: Jenny E. Greene*

**NSF Astronomy & Astrophysics Postdoctoral Fellow (AAPF) at UC San Diego** **July 2023 - July 2026**

*Mentor: Alison L. Coil*

## EDUCATION

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**The University of California, Irvine** **May 2023**

*Doctorate of Philosophy (Ph.D.) in Physics / Advisor: Michael C. Cooper*

*Irvine, CA*

Thesis title: “When Galaxies Go Quiet: Elucidating the Drivers of Environmental Quenching Across Cosmic Time”

**Georgia Institute of Technology**

**May 2017**

*Bachelor of Science (B.S.) in Physics with highest honors / Advisor: Alberto Fernandez-Nieves*

*Atlanta, GA*

## FUNDING & AWARDS

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**Summary:** Total funding awarded (accepted and declined) **\$1,019,850**

### Prize Fellowships

Princeton Future Faculty in the Physical Sciences Fellowship (**\$302,000**)

National Science Foundation – *Constraining the Cosmic Evolution of Environmental Quenching & Predicting Protocluster Populations* (**\$330,000**, Award# 2303800, PI: D. C. Baxter)

The University of California Chancellor’s Postdoctoral Fellowship Program - *Constraining the Cosmic Evolution of Environmental Quenching* (**\$9,850**)

Sullivan Prize Postdoctoral Fellowship (**\$240,000**, Declined)

### Grants

National Science Foundation – *23rd Annual Symposium of the NSF Astronomy and Astrophysics Postdoctoral Fellows* (**\$50,000**, Award# 2437597, PI: J. Moreno, co-PI: D. C. Baxter)

### Other Honors & Awards

UC San Diego Bouchet Honor Society (**\$500**) 2026

Honorable Mention: UC Irvine President’s Dissertation Year Fellowship (**\$1,000**) 2022

AAS National Osterbrock Leadership Program Fellowship 2020-2023

UCI Physics & Astronomy Graduate Student Mentor Award 2020-2023

LSSTC Data Science Fellowship 2019-2022

Honorable Mention: Ford Foundation Predoctoral Fellowship 2019

UCI Eugene Cota-Robles Graduate Fellowship (**\$67,000**) 2017-2023

Tulsa Community Foundation: AMC Cares Scholarship (**\$5,000**) 2015-2017

Georgia Hope Scholarship (**\$15,000**) 2015-2017

## PUBLICATION STATISTICS (FULL LIST ATTACHED AT THE END)

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Peer-reviewed papers: **18** (6 first-author, **12** contributing author)

Citations: **230+**; h-index=**9**

## TEACHING EXPERIENCE

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**Summary:** Taught **6** courses (**1** Instructor of Record, **3** Teaching Assistant, **2** Guest Lecturer) and **12** workshops.

### Instructor of Record

ASTR 20A (Introduction to Astrophysics I) at UC San Diego

Fall 2025

### Teaching Assistant

Physics 20E (Life in the Universe) at UC Irvine

Spring 2019

Physics 20B (Cosmology) at UC Irvine

Winter 2019

Physics 20D (Space Science) at UC Irvine

Fall 2018

### Guest Lecturer

ASTR 2 (Galaxies and the Universe) at UC San Diego

May 2025

Physics 138 (Astrophysics of Galaxies) at UC Irvine

Nov 2020

### Workshops & Bootcamps

STARTastro Python Workshop at UC San Diego

2024-2026

Computational Astrophysics Research Preparation (CARP) Python Workshop at UC San Diego

2024-2026

Computational Research Access NETwork (CRANE) Astronomy Data Analysis Workshop

2023-2025

UC Carpentries Fall Workshop Series at UC San Diego

Sep 2023

LSSTC Data Science Fellowship Program Workshop at University of Washington

June 2023

Physics Olympiad Bootcamp at Ardent Academy

Dec 2019

### Training & Certificates

Instructor Certification from *The Carpentries*

2024

*The Carpentries* is an international organization that teaches coding and data science skills through hands-on workshops. This certificate **recognizes the completion of pedagogical training to teach these skills to a global population of learners.**

Certification in Course Design

2020

This certificate from UCI Division of Teaching Excellence and Innovation **recognizes the completion of training to design effective, inclusive, and engaging courses.**

Certification for the Integration of Research, Teaching and Learning

2020

This certificate from UCI Division of Teaching Excellence and Innovation **recognizes the completion of training to integrate evidence-based teaching practices to enhance student outcomes.**

Mentoring Excellence Program Certification

2019

This certificate from UCI Graduate Division **demonstrates acquisition of mentoring skills, such as cultural competence and inclusivity, effective communication, navigating power dynamics, and conflict mediation.**

## PRESENTATIONS

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**Summary:** **33** presentations (**19** invited, **14** contributed) / †=invited; \*=contributed

### Colloquia/Seminars

University of Washington Colloquium†

April 2026

University of Southern California CosmoLab Seminar†

Mar 2026

Cal State Long Beach Colloquium†

Mar 2026

UC San Diego Community and Science Advancements in Spanish (CaSAS) Seminar†

Sep 2025

Caltech Tea Talk†

May 2025

University of Wisconsin-Madison Science Seminar†

April 2025

UC Irvine Astronomy Seminar†

April 2025

UC Riverside Astronomy Seminar†

April 2025

UC San Diego Simulations TheoRy AND more (STRAND) Meeting†

March 2025

Washington State University Physics & Astronomy Colloquium† [video]

Nov 2024

|                                                                                           |            |
|-------------------------------------------------------------------------------------------|------------|
| University of Florida Astronomy Colloquium <sup>†</sup>                                   | Nov 2024   |
| UC Davis Cosmology and Astronomy Seminar <sup>†</sup>                                     | Oct 2024   |
| Georgia Institute of Technology Center for Relativistic Astrophysics Seminar <sup>†</sup> | Nov 2023   |
| UC San Diego Astronomy Colloquium <sup>†</sup>                                            | Oct 2023   |
| University of Kansas Astronomy Seminar <sup>†</sup>                                       | April 2022 |
| The Ohio State University Little Galaxies Journal Club <sup>†</sup>                       | March 2021 |

### Conferences/Workshops

|                                                                                           |            |
|-------------------------------------------------------------------------------------------|------------|
| GalScale2026*   Ascona, Switzerland                                                       | July 2026  |
| NSF AAPF Fellows Symposium*   Phoenix, AZ                                                 | Jan 2026   |
| Galaxy Formation & Evolution in Southern California 2025*   San Diego, CA                 | Sep 2025   |
| The Baryon Cycle in the Cosmic Web*   Bern, Switzerland                                   | Sep 2025   |
| Tracing Cosmic Evolution with Galaxy Clusters V*   Sesto, Italy                           | July 2025  |
| President's Postdoctoral Fellowship Program Academic Spring Retreat*   Lake Arrowhead, CA | April 2025 |
| Conference for Emerging Black Academics in STEM*   Pasadena, CA                           | April 2025 |
| Ensenada San Diego Astrophysical Society Meeting*   Ensenada, Mexico                      | Sep 2024   |
| Galaxy Formation & Evolution in Southern California 2024*   Pasadena, CA                  | Sep 2024   |
| President's Postdoctoral Fellowship Program Academic Spring Retreat*   Lake Arrowhead, CA | April 2024 |
| AAS 243rd Meeting (Dissertation Talk)*   New Orleans, LA                                  | Jan 2024   |
| NSF AAPF Fellows Symposium*   New Orleans, LA                                             | Jan 2024   |
| The Baryon Cycle of Protocluster Galaxies <sup>†</sup>   Leiden, Netherlands              | Oct 2023   |
| First Structures in the Universe 2023 <sup>†</sup>   Paris, France                        | Sept 2023  |
| Inaugural Conference for Emerging Black Academics in STEM <sup>†</sup>   Pasadena, CA     | May 2023   |
| AAS 240th Meeting*   Pasadena, CA                                                         | June 2022  |
| Georgia Tech College of Science REU Poster Symposium*   Atlanta, GA                       | July 2017  |

## ADVISING & MENTORSHIP

**Summary:** Advised **2** students; mentored **21** students (**4** graduate, **17** undergraduate).

### Research Advising

|                                                                            |              |
|----------------------------------------------------------------------------|--------------|
| Sophia Um (undergraduate), Senior thesis on dwarf galaxies at UC San Diego | 2024-2026    |
| Maxwell Beckman (undergraduate)                                            | 2026-present |

### Academic Mentorship

|                                                                                          |              |
|------------------------------------------------------------------------------------------|--------------|
| UC San Diego Astronomy & Astrophysics Postdoc × Grad Mentorship Program   <b>2</b> grads | 2024-present |
| Computational Astrophysics Research Preparation Program   <b>17</b> undergrads           | 2024-present |
| UC Irvine PACE Mentorship Program Peer Mentor   <b>2</b> grads                           | 2019-2021    |

## SERVICE & LEADERSHIP

**Summary:** Organized **3** conferences/workshops; Served on **2** departmental/student committees; Reviewer for **3** grant/telescope proposals; Developed and/or led **3** mentoring/outreach programs.

### Scientific Organizing

|                                                                                        |      |
|----------------------------------------------------------------------------------------|------|
| Galaxy Formation & Evolution in Southern California Conference                         | 2025 |
| 23rd Annual NSF AAPF Postdoc Symposium                                                 | 2024 |
| “Cultivating the Next Generation of Leaders” Splinter Session at the 240th AAS Meeting | 2022 |

### Departmental Committees

|                                                           |              |
|-----------------------------------------------------------|--------------|
| UCSD Astronomy & Astrophysics Colloquium Committee Member | 2024-present |
|-----------------------------------------------------------|--------------|

### Grant & Telescope Proposal Review

|                                                                                                 |      |
|-------------------------------------------------------------------------------------------------|------|
| Hubble Space Telescope Cycle 33 TAC Expert Reviewer                                             | 2025 |
| Future Investigators in NASA Earth and Space Science and Technology (FINESST) No-Panel Reviewer | 2025 |
| Hubble Space Telescope Cycle 32 TAC Expert Reviewer                                             | 2024 |

Academic Leadership

|                                                                                                                                                                                             |              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Founder and Program Director</b> - Computational Astrophysics Research Preparation (CARP)<br>NSF-funded coding and mentorship program for community college transfer students.           | 2023-present |
| <b>Program Coordinator</b> - Computational Research Access NEtwork (CRANE)<br>International coding and mentorship program for historically underrepresented scholars.                       | 2022-2025    |
| <b>Co-Chair</b> - UCI Physics & Astronomy Community Excellence (PACE) Peer Mentoring Program<br>Student-led program focused on 1-on-1 and group mentoring for first-year graduate students. | 2020-2023    |
| <b>Social Event Chair</b> - UC Irvine Physics Graduate Caucus                                                                                                                               | 2018-2020    |

Outreach

|                                                                                       |              |
|---------------------------------------------------------------------------------------|--------------|
| Invited Speaker - Cosmos & Coffee Astronomy Talk Series                               | May 2026     |
| Instructor - AstroReach Astronomy Essentials Course at The Preuss School              | Jan 2026     |
| Invited Speaker - Peninsula Astronomical Society Talk Series                          | Jan 2026     |
| Invited Speaker - Inclusive Pedagogy Session at Choir Conference                      | July 2025    |
| Invited Speaker - Lick Observatory Summer Series Public Outreach Talks [video]        | July 2025    |
| Invited Speaker - AstroReach San Diego                                                | Feb 2025     |
| Volunteer - UC San Diego Cosmic Tours Portable Planetarium Shows                      | 2025-present |
| Invited Guest - Astro[sound]bites Podcast Episode on Machine Learning in Astrophysics | Oct 2020     |
| Volunteer - UC Irvine Telescope Outreach Program                                      | 2019-2020    |

OBSERVING EXPERIENCE

Summary: 17.5 nights on Keck (KCWI, KCRM, MOSFIRE, DEIMOS) as Co-Investigator across 5 programs.

| Observatory / Instrument | Time       | Program                                                    | Role |
|--------------------------|------------|------------------------------------------------------------|------|
| Keck/KWCI+KCRM           | 1.5 night  | Illuminating the Physics of the Multiphase Baryon Cycle    | Co-I |
| Keck/MOSFIRE             | 1.5 nights | MOSFIRE Survey of Galaxies in Early Rich Environments      | Co-I |
| Keck/KCWI                | 4.5 nights | KCWI Survey of Ultradiffuse Galaxies in the Field          | Co-I |
| Keck/DEIMOS              | 3.5 nights | DEIMOS Survey of the JWST CEERS Field                      | Co-I |
| Keck/DEIMOS              | 6.5 nights | Constraining the Physics of Satellite Quenching at $z < 1$ | Co-I |

MEDIA COVERAGE

|                                                                                 |            |
|---------------------------------------------------------------------------------|------------|
| Seven Scholars Honored for Academic Excellence and Advocacy, UC San Diego Today | April 2026 |
| Black in Galaxy Astrophysics, Nature Astronomy                                  | June 2025  |
| Devontae Baxter, Star Detective, UC Irvine Physical Sciences Communications     | Feb 2021   |
| Some Granular Columns Weigh Too Much, American Physical Society News            | March 2020 |

PROFESSIONAL MEMBERSHIPS

|                                      |              |
|--------------------------------------|--------------|
| American Astronomical Society        | 2020-present |
| National Society of Black Physicists | 2024-present |

SKILLS

Technical Skills: Python, L<sup>A</sup>T<sub>E</sub>X, Unix/Bash, Mathematica, Git, ADQL/SQL, Astrostatistics, Machine Learning  
Soft Skills: Scientific communication, project management, student-centered teaching, mentoring across differences  
Foreign Languages: Spanish (fluent), French (intermediate)

REFERENCES

|                                                                        |                                                                           |                                                                                  |                                                                                |
|------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Reference 1<br>Prof. Michael Cooper<br>cooper@uci.edu<br>Ph.D. Advisor | Reference 2<br>Prof. Alison Coil<br>acoil@ucsd.edu<br>Postdoctoral Mentor | Reference 3<br>Prof. Gregory Rudnick<br>grudnick@ku.edu<br>Research Collaborator | Reference 4<br>Prof. Ethan Nadler<br>enadler@ucsd.edu<br>Research Collaborator |
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# List of Publications

## PEER-REVIEWED PUBLICATIONS

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Summary: **18 publications; 230+ citations; h-index=9** (see ADS library or Google Scholar)

### *First-author publications (6 papers, 80+ citations):*

6. **D. C. Baxter** et al., 2026, “What Becomes of *JWST*/NIRCam-selected High-Redshift Massive Galaxies?” *ApJ*, submitted
5. **D. C. Baxter** et al., 2025b, “Quantifying the Impact of Observational Incompleteness on Identifying and Interpreting Galaxy Protocluster Populations with the TNG-Cluster Simulation” *ApJ*, 990 225
4. **D. C. Baxter** et al., 2025a, “The Importance of Gas Starvation in Driving Satellite Quenching in Galaxy Groups at  $z \sim 0.8$ ” *ApJ*, 979, 41
3. **D. C. Baxter**, et al. 2023, “When the Well Runs Dry: Modelling Environmental Quenching of High-mass Satellites in Massive Clusters at  $z \gtrsim 1$ ” *MNRAS*, 526, 3716
2. **D. C. Baxter**, et al. 2022, “The GOGREEN Survey: Constraining the Satellite Quenching Timescale in Massive Clusters at  $z \gtrsim 1$ ” *MNRAS*, 515, 5479
1. **D. C. Baxter**, et al. 2021, “A Machine Learning Approach to Measuring the Quenched Fraction of Low-Mass Satellites Beyond the Local Group” *MNRAS*, 503, 1636

### *Co-authored publications (12 papers; contributed to data acquisition/analysis and/or writing):*

12. S. Um, **D. C. Baxter**, E. Nadler to be submitted., “Linking Orbital History to the Quenching of Isolated Dwarf Galaxies”
11. F. J. Mercado, **D. C. Baxter**, M. K. Rodriguez Wimberly, et al., 2026, “The Quenched Fraction of Satellites Around Simulated Milky Way-mass Galaxies” *ApJ*, 1004, 217
10. D. Sikorski, et al. (incl. **D. C. Baxter**), 2026, “The *HST*-Hyperion Survey: Environmental Imprints on the Stellar-Mass Function at  $z \sim 2.5$ ” *A&A*, 708, A114
9. F. Giddings, et al. (incl. **D. C. Baxter**), 2026, “The *HST*-Hyperion Survey: Companion Fraction and Overdensity in a  $z \sim 2.5$  Proto-supercluster” *A&A*, 706, A55
8. L. Sandoval Ascencio, et al. (incl. **D. C. Baxter**), 2025, “Caught in the Act of Quenching: A Population of Post-Starburst UDGs” *OJAp*, 8, 110
7. G. Hewitt, et al. (incl. **D. C. Baxter**), 2025, “Distinct Origins of Environmentally Quenched Galaxies in the Core and Virialized Regions of Massive Clusters at  $0.8 < z < 1.5$ ” *MNRAS*, 541, 49
6. H. Gully, et al. (incl. **D. C. Baxter**), 2025, “Insights into Environmental Quenching at  $z \sim 1$ : An Enhancement of Faint, Low-mass Passive Galaxies in Clusters” *MNRAS*, 539, 3058
5. G. Gururajan, et al. (incl. **D. C. Baxter**), 2025, “Gas Properties as a Function of Environment in the Proto-supercluster Hyperion at  $z \sim 2.45$ ” *A&A*, 698, A312
4. L. Xie, G., et al. (incl. **D. C. Baxter**), 2024, “The First Quenched Galaxies: When and How?” *ApJL*, 967, L42
3. E. Kukstas, et al. (incl. **D. C. Baxter**), 2023, “The GOGREEN Survey: A Critical Assessment of Environmental Trends in Cosmological Hydrodynamical Simulations at  $z \approx 1$ ” *MNRAS*, 518, 4782
2. M. K. Rodriguez Wimberly, M. C. Cooper, **D. C. Baxter**, et al., 2022, “Sizing from the Smallest Scales: The Mass of the Milky Way” *MNRAS*, 513, 4968
1. S. Mahajan, M. Tennenbaum, S. Pathak, **D. C. Baxter**, et al., 2020, “Reverse Janssen Effect in Narrow Granular Columns” *PRL*, 124, 128002

### *Other Publications (1 white paper):*

1. The Fellows of the National Osterbrock Leadership Program (incl. **D. C. Baxter**), 2023, “Reimagining the Astronomy PhD for the 21st Century” white paper, *American Astronomical Society*